

Price-Path Normality Audit for Wilder’s Log-Odds RSI

Multiplicative price paths, scale calibration, and residual shape

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Reproducible computational supplement

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Abstract

This computational supplement asks whether the normalization formulas derived for additive i.i.d. innovations remain useful for classical RSI when the input is a multiplicative price path. For each experiment, symmetric i.i.d. log returns are generated, exponentiated into a positive price series, and then converted to ordinary arithmetic price changes before Wilder smoothing. The same innovation path is also processed as log-price increments, which is the additive benchmark covered directly by the manuscript. The audit contains six innovation laws, five volatility levels, twelve lookbacks, and three million retained observations per law–volatility pair. At small log-return volatility the price and log-price coordinates are close. As volatility and lookback increase, the classical price coordinate develops positive skewness that variance normalization cannot remove. The manuscript normalizers remain useful scale corrections, but they are not exact pivots for the multiplicative price model and do not imply finite-sample normality.

1 Question and scope

The manuscript studies a generic additive input

$$X_t = X_{t-1} + \varepsilon_t. \quad (1)$$

If $X_t = \log P_t$, the innovations are log returns and the interpretation is direct. Classical market RSI is usually computed from P_t itself. Under

$$P_t = P_{t-1}e^{r_t}, \quad (2)$$

its input increments are

$$\Delta P_t = P_{t-1}(e^{r_t} - 1), \quad (3)$$

which are state dependent and asymmetric even when r_t is symmetric. The question is therefore numerical: how accurately do the additive-model normalizers describe the classical price-RSI coordinate after this nonlinear mapping?

This is a synthetic robustness audit, not a proof for geometric price models, not a backtest, and not evidence that market returns follow any law used here. The deterministic RSI and endpoint identities remain exact on every generated path. Only the probabilistic calibration is under examination.

2 Experimental design

For every law and volatility level, the audit generates 3,000,000 retained observations after a burn-in of $32 \times 377 = 12,064$. The initial price is 10,000; a single global rescaling is applied internally to avoid floating-point overflow and cannot change RSI. The fixed seed is 20260925.

The laws are Gaussian, Laplace, and standardized Student t_ν with $\nu \in \{4, 5, 6, 8\}$. Every innovation has variance one before multiplication by a log-return standard deviation in

$$\{0.5\%, 1\%, 2\%, 4\%, 8\%\}. \quad (4)$$

The lookback grid is

$$n \in \{2, 3, 5, 8, 13, 21, 35, 55, 89, 144, 233, 377\}. \quad (5)$$

Gaussian log returns generate lognormal one-step price ratios. The Laplace and Student cases are alternative multiplicative stress models, not geometric Brownian motion. In particular, untruncated Student log returns have no positive exponential moment; those cases are finite-sample stress experiments and are not covered by the manuscript’s light-tail expansion.

For each path two coordinates are calculated:

$$L_{t,n}^P = \log \frac{U_{t,n}^P}{D_{t,n}^P}, \quad U^P, D^P \text{ smooth } (\Delta P_t)_+, (-\Delta P_t)_+, \quad (6)$$

$$L_{t,n}^{\log P} = \log \frac{U_{t,n}^r}{D_{t,n}^r}, \quad U^r, D^r \text{ smooth } (r_t)_+, (-r_t)_+. \quad (7)$$

The latter is the exact additive benchmark. Each coordinate is viewed under the asymptotic scale $\sqrt{n}/C_F$, the law-specific finite- n correction, its centered version, and an empirical-variance oracle. Centering isolates shape from a finite-run location error; the oracle is diagnostic and is not a proposed usable normalizer.

3 Diagnostics and computational validation

The diagnostics are variance, skewness, excess kurtosis, Kolmogorov distance from $N(0,1)$, central coverage, tail frequency, and Q–Q deviations. For the log-price coordinate, exact scale invariance makes the results identical across the five volatility labels; those labels are therefore not independent replications.

The computational validation confirms:

- all 2,880 expected summary rows are present with unique keys and finite required metrics;
- the reported variances satisfy the algebraic scale identities to a maximum absolute discrepancy of 1.33×10^{-15} ;
- SciPy filtering agrees with a direct Wilder recursion;
- the generated price path reconstructs the supplied log returns and is invariant to a global price rescaling;
- the endpoint log-odds identity holds numerically; and
- price RSI converges to log-price RSI in the small-volatility limit.

These checks assess implementation correctness, completeness, algebraic consistency, and simulation stability; they are not a formal proof of the multiplicative model. Aggregate outputs and representative draws are provided in the accompanying reproducibility package; see Appendix B.

4 Results

4.1 Price versus log-price benchmark

Table 1 averages Kolmogorov distance over all six laws under the centered law-specific correction. The log-price values are invariant to the volatility multiplier, as homogeneity requires. The price values increase approximately with volatility and are more sensitive at long lookbacks.

Table 1: Mean Kolmogorov distance over six laws.

σ_r	n	log-price RSI	price RSI
0.5%	55	0.0012	0.0037
0.5%	377	0.0036	0.0110
1.0%	55	0.0012	0.0072
1.0%	377	0.0036	0.0199
2.0%	55	0.0012	0.0141
2.0%	377	0.0036	0.0382
4.0%	55	0.0012	0.0282
4.0%	377	0.0036	0.0752
8.0%	55	0.0012	0.0564
8.0%	377	0.0036	0.1704

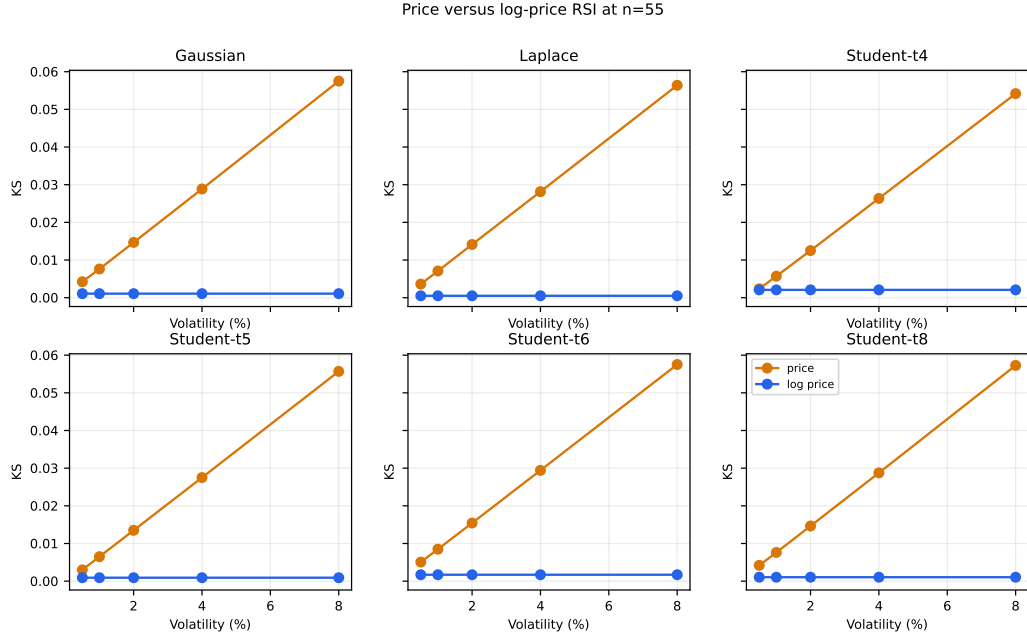
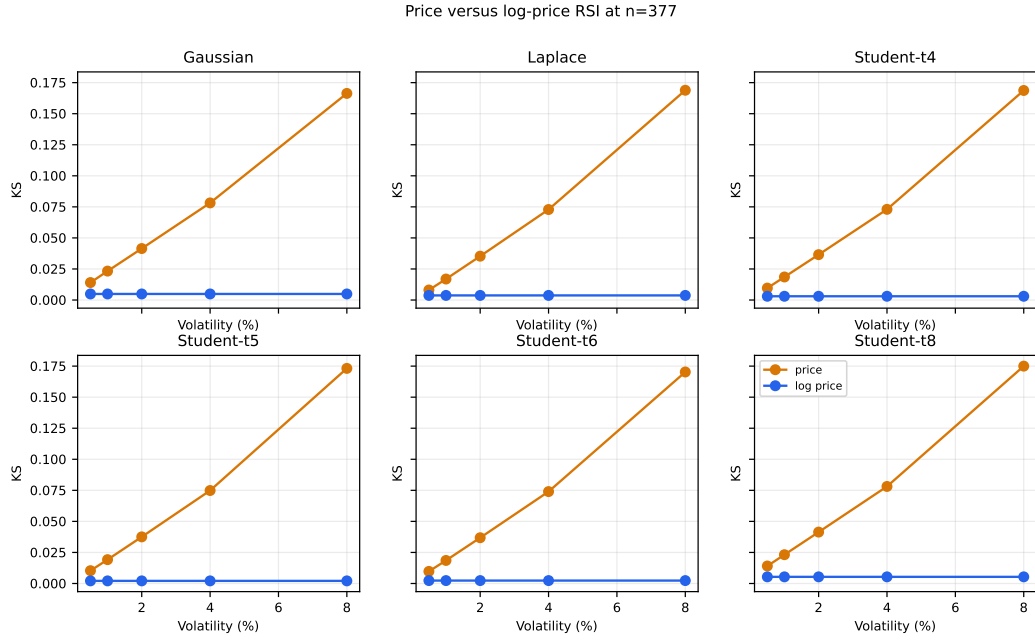
(a) Lookback $n = 55$.(b) Lookback $n = 377$.

Figure 1: Classical price RSI versus the log-price benchmark. Long memory accumulates the multiplicative asymmetry more visibly.

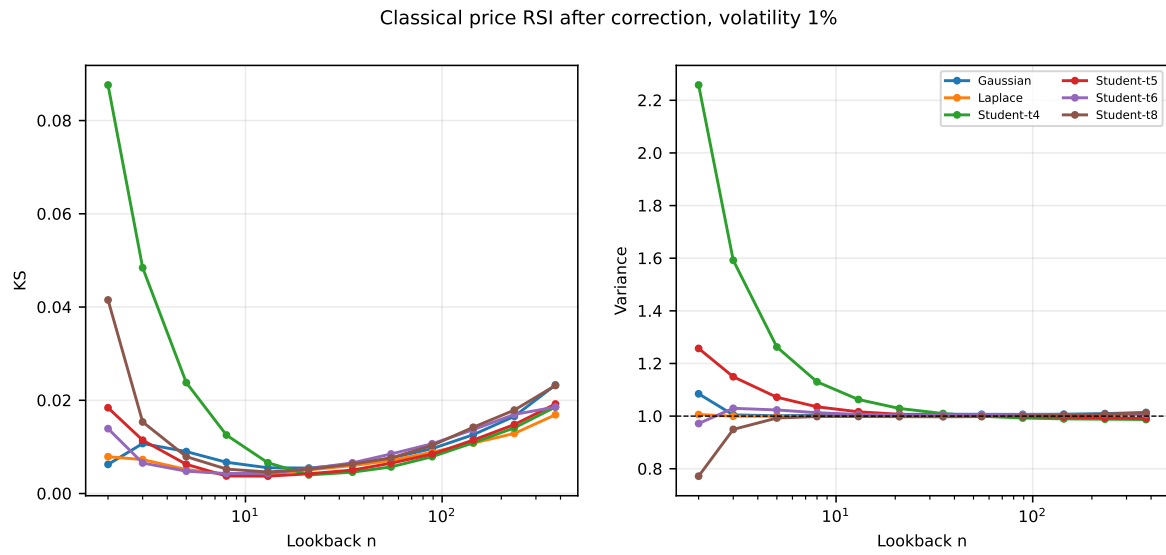
4.2 Scale correction and residual shape

The corrected variances in Table 2 remain near one in most settings, including cases where Kolmogorov distance becomes material. Thus $g_F(n)$ can correct scale well while the resulting variable remains non-Gaussian because of skewness and tail shape.

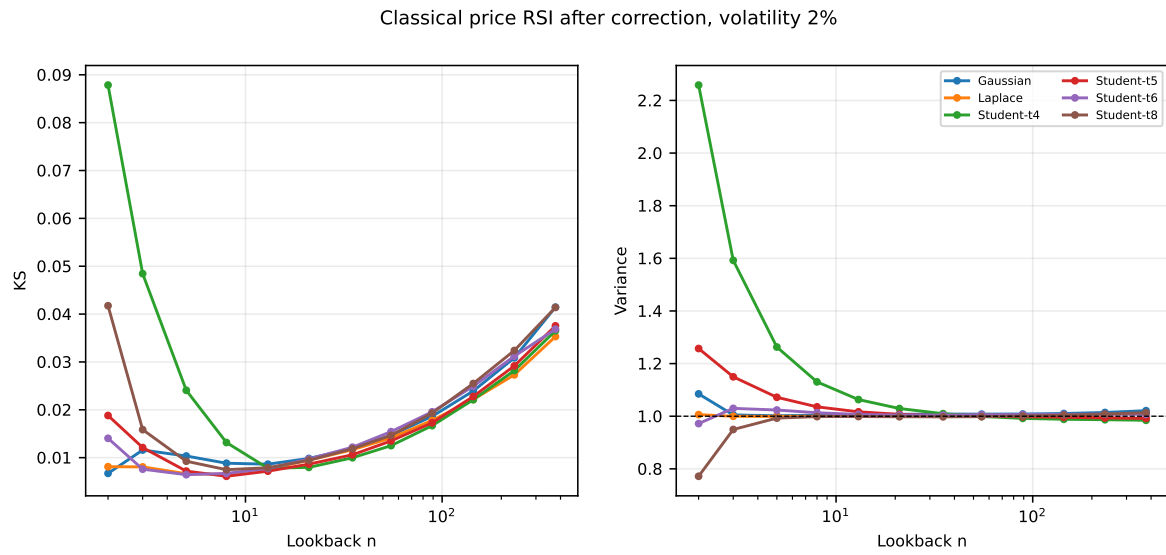
Table 2: Selected classical price-RSI diagnostics under the centered law-specific correction. Tail is $\Pr(|Z| > 3)$.

Law	σ_r	n	Var	Skew	Excess kurt.	KS	Tail
Gaussian	1%	55	1.007	0.120	0.054	0.0076	0.0031
Gaussian	1%	377	1.014	0.318	0.127	0.0233	0.0041
Gaussian	2%	55	1.008	0.228	0.112	0.0147	0.0036
Gaussian	2%	377	1.021	0.595	0.521	0.0414	0.0067
Gaussian	8%	55	1.020	0.879	1.226	0.0575	0.0096
Gaussian	8%	377	1.069	2.264	7.955	0.1663	0.0212
Laplace	1%	55	1.004	0.103	0.040	0.0071	0.0031
Laplace	1%	377	0.989	0.256	0.116	0.0169	0.0034
Laplace	2%	55	1.004	0.209	0.088	0.0141	0.0035
Laplace	2%	377	0.989	0.538	0.463	0.0353	0.0058
Laplace	8%	55	1.009	0.844	1.089	0.0564	0.0091
Laplace	8%	377	1.005	2.244	7.766	0.1689	0.0193
Student- t_4	1%	55	0.999	0.071	0.056	0.0057	0.0031
Student- t_4	1%	377	0.987	0.238	0.066	0.0186	0.0030
Student- t_4	2%	55	0.999	0.177	0.089	0.0125	0.0034
Student- t_4	2%	377	0.985	0.506	0.341	0.0365	0.0051
Student- t_4	8%	55	1.000	0.814	1.044	0.0542	0.0084
Student- t_4	8%	377	0.974	2.098	6.414	0.1687	0.0182

At 0.5–1% volatility, discrepancies are small through much of the lookback grid. At 2%, the Gaussian price coordinate has KS 0.0147 at $n = 55$ and 0.0414 at $n = 377$. At 8% and $n = 377$, the laws have KS near 0.17 and pronounced positive skewness. Those are shape departures, not failures of the variance formula alone.



(a) Log-return volatility 1%.



(b) Log-return volatility 2%.

Figure 2: Corrected classical price-RSI diagnostics at moderate log-return volatility.

4.3 The $n = 2$ stress point

The smallest lookback is deliberately retained. It shows that an asymptotic or first-shift normalizer need not be accurate merely because the innovations are standardized. For Student- t_4 , the corrected scale equals the asymptotic scale ($K_4 = 0$), giving variance about 2.26 and a 4.84% frequency beyond three standard-normal units.

Table 3: Classical price-RSI diagnostics at $n = 2$ and $\sigma_r = 1\%$ under the centered law-specific correction.

Law	Var	Excess kurt.	KS	$\Pr(Z > 3)$
Gaussian	1.085	0.666	0.0062	0.0083
Laplace	1.006	0.525	0.0079	0.0059
Student- t_4	2.258	0.606	0.0876	0.0484
Student- t_5	1.257	0.625	0.0184	0.0121
Student- t_6	0.972	0.638	0.0139	0.0057
Student- t_8	0.772	0.632	0.0415	0.0026

For Student- t_4 , the reported excess kurtosis is a finite-sample diagnostic; the innovation law has no finite population fourth moment.

5 Limitations

- The study uses stationary i.i.d. synthetic log returns. It does not test serial dependence, heteroskedasticity, regime changes, or empirical market data.
- The additive-model normalizers are theorems for increments of the chosen additive input. Their performance for classical price RSI under multiplicative dynamics is an empirical robustness result, not a new theorem.
- The retained coordinate paths are serially dependent. The reported distributional distances are descriptive Monte Carlo estimates rather than i.i.d. test statistics.
- Untruncated Student log returns have no positive exponential moment and are used here only as finite-sample stress experiments. In addition, Student t_4 has no finite population fourth moment, so its reported excess kurtosis is a sample diagnostic.
- The audit does not identify a return law for financial data. Gaussian, Laplace, and Student innovations are controlled inputs used to separate scale effects from residual shape effects.

6 Conclusion

The audit establishes a sharp distinction between exact applicability and empirical robustness. When the RSI input is log price and its increments follow the assumed law, the additive-model normalizers apply directly. For classical price RSI, they remain useful small-return approximations: at low and moderate synthetic volatility, much of the scale correction survives the exponential mapping.

Variance calibration is not, however, a proof of normality. The multiplicative price map introduces state dependence and positive skewness that become more visible as volatility and lookback increase. The formulas should therefore be described as exact or asymptotic results for additive increments and as volatility-dependent benchmarks for ordinary price RSI. The Student first-shift corrections come from the companion conventional technical note and remain outside the Lean-verified scope.

A Selected Q–Q grids

The Q–Q panels superimpose classical price RSI (orange) and the log-price benchmark (blue). At low volatility the curves are close except in stressful short-lookback cases. As volatility increases, the orange curve develops a systematic asymmetric bend while the blue benchmark stays fixed. The first pair contrasts Gaussian inputs at low and high volatility; the second pair shows alternative innovation laws.

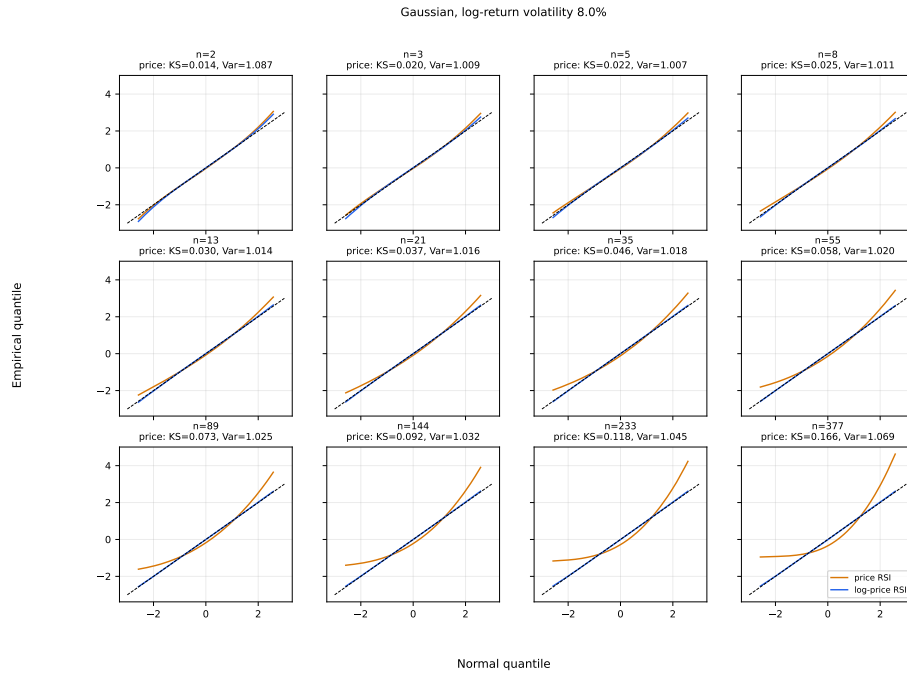
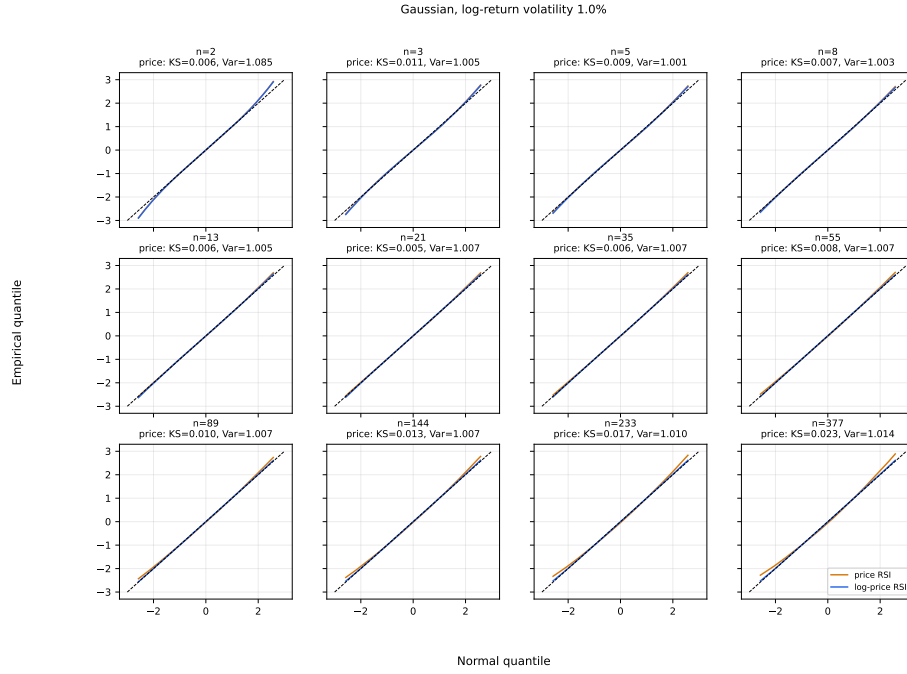


Figure 3: Gaussian Q–Q grids. At 8% volatility, variance remains near one while positive skewness invalidates a normal interpretation.

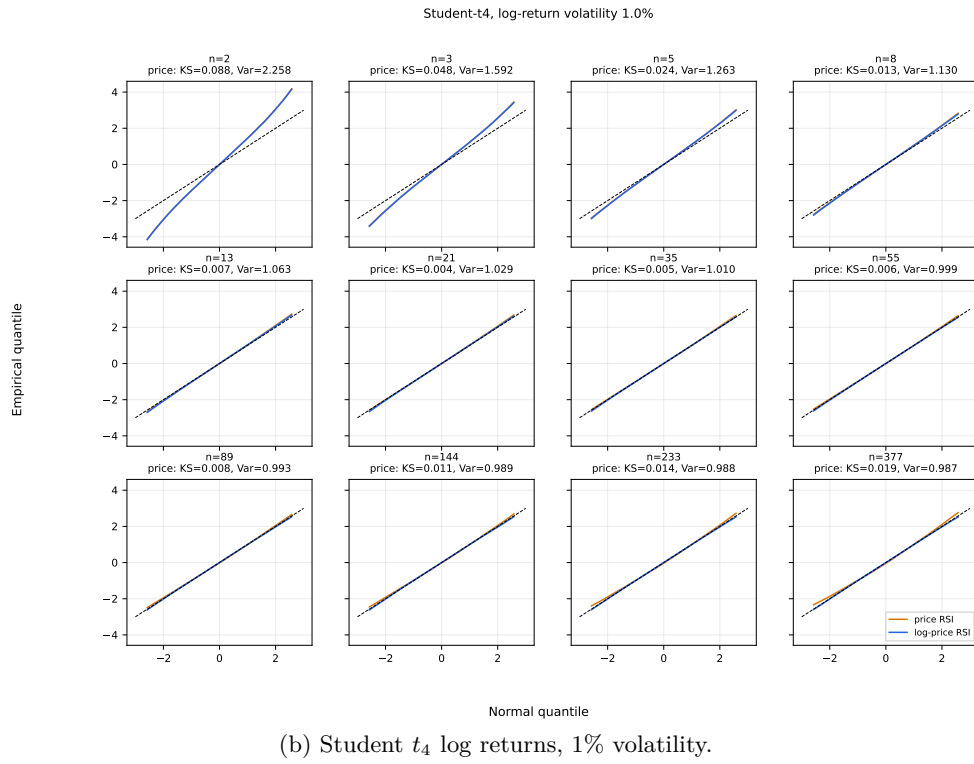
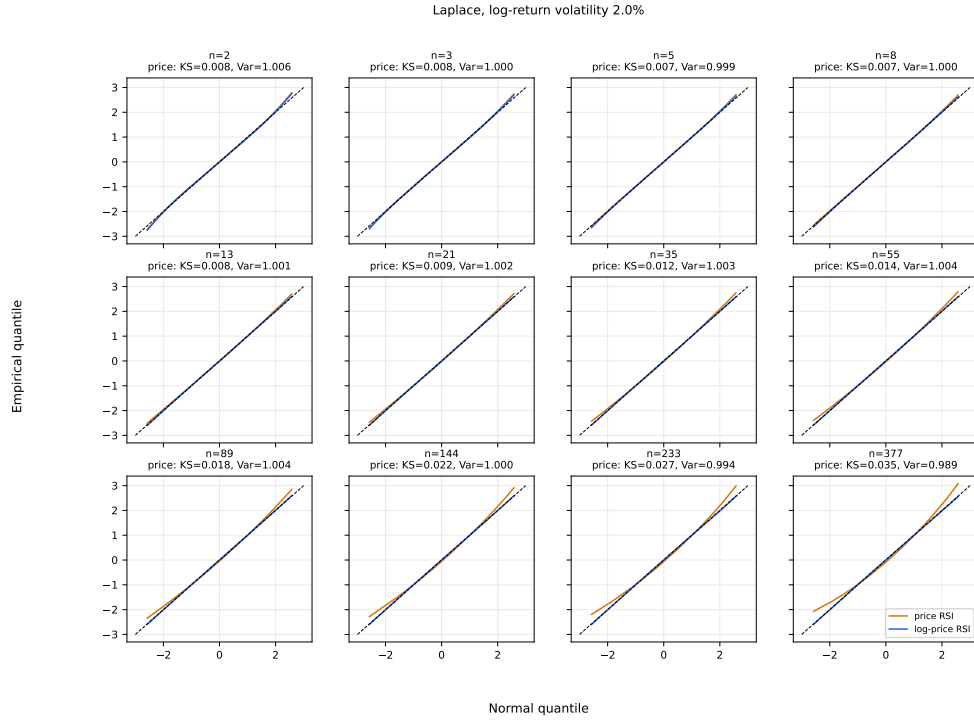


Figure 4: Laplace and Student t_4 Q-Q grids. The Student t_4 scale error at $n = 2$ is inherited from the first-shift approximation, for which $K_4 = 0$.

B Reproducibility and provenance

The archived reproducibility package contains the complete code, pinned Python 3.12 dependencies, aggregate CSV outputs, quantile and empirical-CDF grids, histograms, block estimates, representative draws, figures, metadata, and the implementation-validation script. The log-price rows repeated across the five volatility labels are identical by exact scale invariance and are not independent replications.

On Windows, run from the repository root:

```
py -m pip install -r supplement\price_path_normality_audit\requirements.txt
py supplement\price_path_normality_audit\validate_implementation.py
py supplement\price_path_normality_audit\run_price_path_audit.py
```

On Linux or macOS, use:

```
python3 -m pip install -r supplement/price_path_normality_audit/requirements.txt
python3 supplement/price_path_normality_audit/validate_implementation.py
python3 supplement/price_path_normality_audit/run_price_path_audit.py
```

Published aggregate data and representative draws are sufficient for review without repeating the complete simulation. The `-plots-only` option regenerates the figures from those outputs.

The author formulated the research question, mathematical framework, and interpretation, and takes responsibility for the claims made in this report. AI tools assisted with computational design, code drafting, debugging, visualization, and editorial preparation. Numerical results were produced by the archived code and preserved outputs; they are not Lean-verified theorems.